Raluca D. Gaina

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Game AI developer with 10+ years of experience designing intelligent systems, gameplay simulations, and applied AI solutions for games and interactive systems. Strong background in prototyping and implementing gameplay features in Unity (C#) and Java, with expertise in AI-driven behaviours and systems design. Experienced collaborating with multidisciplinary teams, contributing to game design decisions, and translating complex mechanics into scalable gameplay systems.

Skills

Gameplay Programming

- ✔ Unity (C#), Java, Python
- ✔ Gameplay systems design and implementation
- ✔ Prototyping and rapid iteration
- ✔ Debugging, optimisation, testing
- ✔ Game architecture and maintainable code

AI & Systems

- ✔ AI behaviours and decision systems
- ✔ Reinforcement Learning, Evolutionary Algorithms, Procedural Content Generation
- ✔ Simulation systems, AI-driven testing

Collaboration & Production

- ✔ Cross-disciplinary collaboration (design, QA, etc.)
- ✔ Technical documentation and communication
- ✔ Code reviews and mentoring
- ✔ Git / Version control
- ✔ Translating AI research into practical applications for game design testing

Experience

Co-founder / Director
Tabletop R&D | 2023 – Present

- Developed gameplay simulation systems for AI-driven playtesting of tabletop board games
- Created “digital twins” of physical games to model gameplay mechanics and run thousands of simulations to optimise balance and design
- Prototyped and iterated on gameplay systems with commercial clients including Bright Eye Games, AP Board Games, and Underdog Games
- Secured £120k+ in grant funding to support product development and innovation
- Leading transition toward a SaaS product and user-facing tools
- Defined product direction, technical roadmap, and AI integration strategy
- Communicated with a variety of technical and non-technical stakeholders

Game AI Researcher & Lecturer
Queen Mary University of London | 2021 – 2026

- Designed and implemented gameplay systems and AI behaviours in Unity (C#) and Java for game prototyping and analysis
- Led, designed and delivered MSc and BSc courses in Game AI and game development
- Supervised 20+ student projects, many producing playable prototypes and research outputs
- Regularly communicated complex AI and game concepts to diverse audiences
- PhD supervisor and research impact coordinator in IGGI Doctoral Programme

Research & Teaching Roles (AI & Games)

Queen Mary University of London / University of Essex | 2016 – 2021

- Contributed to multiple AI and game-related research projects
- Collaborated across research and industry partnerships (e.g., DSTL, Alan Turing Institute)
- Developed and delivered teaching in AI, game development, and computational creativity
- Developed and prototyped AI-driven systems in game contexts
- Completed research internship at Microsoft Research working on multi-agent reinforcement learning environments
- Collaborated with interdisciplinary teams across academia and industry

Selected AI & Game Projects

1. **AI Playtesting & Simulation Platform (Tabletop R&D)**. End-to-end system for modelling, simulating, and analysing board games using AI agents. Based on the **Tabletop Games Framework** (https://github.com/GAIGResearch/TabletopGames/), leading innovative research in AI and tabletop board games.
2. **General Video Game AI Research**. Developed planning and decision-making systems using evolutionary algorithms, and led state of the art AI competitions. **Multi-Agent Reinforcement Learning Environments**. Designed and implemented competitive environments for AI benchmarking with Microsoft Research.
3. **Game Development (Unity/C# & Java)**. Built multiple playable game prototypes, implementing gameplay mechanics, AI systems, and core game loops, with a focus on rapid prototyping and iteration. Unity (primary), experience adapting to new engines and tools.

Education

- [2021] PhD, Intelligent Games & Game Intelligence – Queen Mary University of London
- [2016] MSc, Computer Games (Distinction) – University of Essex
- [2015] BSc, Computer Games (First Class) – University of Essex

Selected Leadership & Community

- General Chair, IEEE Conference on Games 2026
- Organiser, London Game AI Meetup (2019–2024)
- Chair and reviewer across major AI & games conferences and journals
- Mentoring roles and part of Athena Swan and Women in Games teams

Selected Achievements

- Best Artifact Award, Best PC Member Nomination – AIIDE 2019
- Best Paper Award Nomination – IEEE Conference on Games
- MSc Project Prize and K.F. Bowden Memorial prize (highest marks) – University of Essex
- Multiple competition placements in AI, game development and science communication challenges, including IEEE Present Around the World and IEEE CoG video competitions

Research Publications

A complete list of publications is available at https://rdgain.github.io/publications